SharePlate: Transforming Food Waste into Hope

Track: Web & App Development

Our Problem: To connect students to meaningful causes through an innovative volunteer & NGO portal.

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The Urgent Problem: Waste Amidst Want

A Global Paradox

Every year, 1.3 billion tons of food is wasted globally, while over 800 million people suffer from hunger. This stark contrast highlights a critical inefficiency in our food systems.

Local Impact: Campus & City

Universities, corporate events, hostels, and restaurants are major contributors to surplus food. Without proper systems, this valuable resource is often discarded, leading to economic and environmental costs.

Gaps in Current Systems

Manual & Unverified Processes: Leading to wasted resources and inefficiency.

Lack of Real-time Tracking

No immediate oversight for food donors, NGOs, or delivery partners.

Trust, Transparency & Scalability

Current methods often lack the accountability needed for broad adoption.

SharePlate directly addresses United Nations SDG Goals: Zero Hunger, Responsible Consumption & Production, and Climate Action.

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SharePlate: Our Innovative Solution

SharePlate is an AI-driven web and mobile platform designed to seamlessly connect food donors, NGOs, and delivery partners.

AI-Powered Recipe Assistant

Maximizes food utilization by suggesting creative recipes and redistribution strategies for surplus ingredients.

Real-Time Food Rescue

Instant, intelligent matching of available food with NGO demand, ensuring rapid allocation.

Verified Delivery

Utilizes QR codes and geolocation for transparent pickup and drop-off, building trust and traceability.

Gamification & CSR Rewards

Incentivizes participation from individuals and businesses through badges, leaderboards, and CSR recognition.

Food Marketplace

Facilitates the distribution of near-expiry food at reduced or zero cost, minimizing waste.

> Uniqueness: Unlike typical NGO portals, SharePlate integrates cutting-edge AI, engaging gamification, and robust cloud scalability into one cohesive ecosystem for maximum impact.

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Our Method: A Seamless Ecosystem

Key Modules in Action

- Donor Module: Users upload food details, photos, quantity, and expiry. AI then suggests optimal redistribution.
- NGO/Beneficiary Matching: AI-driven engine intelligently pairs food supply with real-time NGO demand, prioritizing based on location, urgency, and food type.
- Delivery Verification: QR code scanning at pickup and drop-off, coupled with GPS tracking, ensures accountability and transparency.
- Gamification Layer: Introduces badges, leaderboards, and CSR points to foster social motivation and continuous engagement.

Tech Stack

- Frontend: React.js (Web), Flutter (Mobile)
- Backend: Node.js / Django / Firebase
- Database: Firestore / MongoDB
- AI/ML: TensorFlow Lite / OpenAI APIs (for chatbot & recommendations)
- Security: JWT authentication, role-based access control
- Verification: QR code integration & Google Maps API

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Expected Outcomes: Impact at Every Level

Social Impact

Thousands of meals rescued, directly bridging the critical gap between food waste and hunger, fostering a more equitable community.

Environmental Impact

Significant reduction of CO_2 emissions by diverting food from landfills, where it decomposes and releases harmful greenhouse gases.

Community Engagement

Empowers donors through recognition, strengthens NGOs with resources, and reduces hunger, fostering a cycle of generosity.

Corporate Value

Provides seamless CSR integration, allowing companies to easily adopt SharePlate as a core part of their sustainability initiatives.

Scalability & Growth

Designed with a robust architecture for effortless expansion across multiple cities and countries, ensuring minimal overhead.

Future Scope: Building a Smarter Food Ecosystem

Predictive Analytics

Leveraging data to forecast food surplus patterns, enabling proactive pickups and optimizing logistics.

Blockchain Transparency

Implementing immutable records for all donations, significantly boosting NGO credibility and donor trust.

Expanded Marketplace

Integrating with restaurants and supermarkets for widespread redistribution of near-expiry groceries.

IoT Integration

Deploying smart sensors in kitchens and event venues to automatically report surplus food in real-time.

Policy Partnerships

Collaborating with municipal food safety boards and advocating for CSR tax benefits to incentivize participation.

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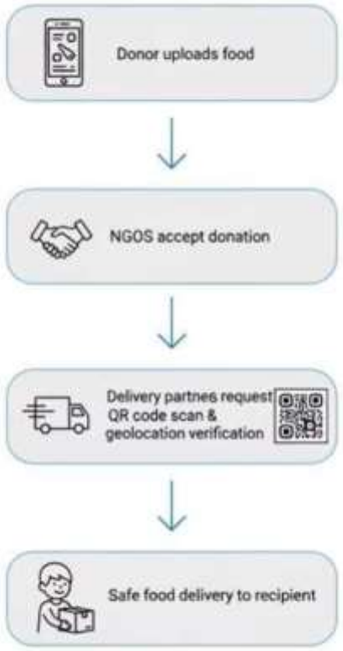
Conclusion: The Driving Force Behind SharePlate

ByTechies: Our Foundation of Excellence

- **Unparalleled Technical Acumen: Our diverse team leverages profound expertise in web development, mobile solutions, cutting-edge AI/ML, and robust cloud infrastructure.**
- **Proven Impact: With a distinguished track record, we deliver real-world solutions across IoT, advanced data analytics, and impactful social good initiatives.**
- **Designed for Scalability & Adaptability: SharePlate is engineered for agile deployment, demonstrating immediate readiness and robust potential for expansive future growth.**

> **"SharePlate transcends technology; it's a powerful statement of community empowerment. We don't just connect surplus to need; we cultivate a sustainable future, one shared meal at a time."**



"SharePlate - Transforming Food Waste into Hope with Technology, AI, and Community Engagement."